Finite terms with the exact $\vee$ dependence: check, DGLAP and running coupling, simplification, final result, and comparison with the final result of arXiv:1411.2869

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1 What is different from the previous version

- **Kinematics.** $\Lambda < p^+ < \vee$, $\Lambda < k^+ < \vee$, with no assumption on how close k^+ (or p^+) is to $\vee$ or to Λ. So

$$a \equiv \frac{k^+}{\vee} \in (0,1)$$

  is kept exactly everywhere. The only limit taken is $\Lambda \to 0$, and only where it exists, i.e. everywhere except in $\log \frac{\vee}{\Lambda}$ (your evolution terms).

- **Removed** from `Finite_terms_final.pdf`, because they needed $k^+/\vee \to 0$:
  - the section "Power corrections" (dropping $\log(1 - \frac{k^+}{\vee})$, $\frac{1}{\vee}$, the $\frac{k^+}{\vee}$ inside R^i, and F6);
  - the split of every finite row into $C\,L_\vee + R$ and the hidden $\mathcal{N} L_\vee (\mathcal{B} - \mathcal{J})$;
  - the region-T/region-P decomposition of Group III Rows III, V, VI;
  - the "reg" in the DGLAP term (the lower limit $\frac{k^+}{\vee}$ is now kept).

- **New** (blue throughout): exact $\Lambda \to 0$ forms of all rows; closed forms of every remaining p^+ integral; the pole tally with $\log(1-a)$ kept, which leaves an uncancelled pole (Sec. 7); the four-ρ finite term (Sec. 9); the final result of arXiv:1411.2869 next to yours, term by term (Secs. 12, 13).

2 Summary

1. Are these the final terms? Yes, with the corrections F1–F7 (Sec. 4). The check compares every row with the exact p^+ integral of its first-principles integrand at $\vee/k^+ = 6$, 40, 300, i.e. it never used $k^+ \ll \vee$. Agreement: 10^{-12} or better for the closed-form rows, and $O(\Lambda/k^+)$ for the rows whose plus prescriptions start at 0. **But** with a kept exactly, F6 is no longer a power correction: the "NA" rows are not zero (Sec. 9).

2. Segregation.

- **Running coupling** (Group I Row II part 1, Row V 2b UV). The β_0 terms cancel exactly, for every a. Left over: $\frac{\alpha_s}{2\pi} N_c \big[\frac{67}{9} - \frac{\pi^2}{3} - \log^2(1-a) - L_\vee \log(1-a)\big]\mathrm{LO}$.
- **DGLAP** (Group II Row IV, lines 1–2). The pole is $\frac{\alpha_s}{2\pi}(-\frac{2}{\epsilon}) \int_{k^+/\vee}^1 \frac{d\xi}{\xi^2} P_{gg}(\xi)\,\mathrm{LO}_\xi$ and goes into $D_{h/g}$ exactly; the lower limit $\frac{k^+}{\vee}$ is the kinematic limit of the parent momentum. The finite part is $-\frac{\alpha_s}{2\pi} \int_{k^+/\vee}^1 \frac{d\xi}{\xi^2} P^{(r)}_{gg,+}\,\mathrm{LO}_\xi\big[\ell_\Delta(\mu_F)\big]$.

3. Two results that appear only because a is kept exactly.

1. **A pole is left over:** $\frac{\alpha_s}{2\pi}\frac{2}{\epsilon} N_c \log(1 - \frac{k^+}{\vee})\,\mathrm{LO}$. It comes from the $\log(\frac{\vee}{k^+} - 1)$ of Group I Row II part 1. It depends on $k^+/\vee$, not on ξ, so it cannot go into $D_{h/g}$, and no other row cancels it (`sympy`, Sec. 7). In AABKL the virtual term has no such limit. Question Q1.
2. **The four-ρ rows leave a finite term**, $\mathcal{N} \log(1 - \frac{k^+}{\vee})\times$ (the four-ρ lines with upper limit $\vee - k^+$). Their symmetrized parts do not cancel among themselves. Question Q2.

4. Simplification (Sec. 8).

- Every row is written with $\Lambda = 0$ and the exact a.
- Every remaining p^+ integral is done in closed form with the exponential integral E_1. This includes Group I Row V and Group III Rows III, V, VI; for Row V part 2 the closed forms are written through master integrals.
- All forms were checked against your expressions at $\vee/k^+ = 1.3$, 2.5, 10, 300.
- The exact relations of the three-ρ terms hold for every a: Group I Row IV + Group III Row I = 0; Group II Row I part 1 (A) + Group III Row II collapse into one term.
- The rest of the three-ρ terms do not cancel.

5. Final result: Eq. (11.1).

6. Comparison with AABKL (Secs. 12, 13). Their final answer (Eqs. (4.20), (4.22)) and yours are written next to each other.

- **The same in both:**
  - the $g \to gg$ splitting function and its plus prescription;
  - the kinematic limit ($\int_0^{1-x_p} \leftrightarrow \int_{k^+/\vee}^1$);
  - the after-target (fragmentation) term, including the scheme constant ($1.26 = c_0^2$);
  - the hadron convolution ($\int_{x_F}^1 \frac{d\zeta}{\zeta^2} \leftrightarrow \int_{k^+/\vee}^1 \frac{d\zeta}{\zeta^3}$);
  - the colour topologies of the before-target, interference and virtual terms.
- **Different:** everything that comes from the projectile being colour charges rather than a collinear gluon:
  - the order of LO;
  - the transverse kernels;
  - no PDF subtraction;
  - linear polarization;
  - emission after the target;
  - the running-coupling block;
  - the three- and four-ρ terms;
  - and, as regularization, the Ioffe time versus the p^+ cutoffs.

7. Questions: Sec. 14.

3 Notation

- $a \equiv \frac{k^+}{\vee}$, $L_\vee \equiv \log\frac{\vee}{k^+} = -\log a$. $L_\vee$ is only a name; nothing is assumed about its size.
- $\ell_r \equiv \log\frac{r^2\mu^2_{\overline{MS}}}{c_0^2} = 2\gamma_E + \log\frac{r^2\mu^2_{\overline{MS}}}{4}$, $c_0 = 2e^{-\gamma_E}$, $\hat{E}_r \equiv \frac{2}{\epsilon} + \ell_r$.
- $A \equiv (x-z)^2$, $B \equiv (x-w)^2$, $A' \equiv (x'-z)^2$, $B' \equiv (x'-w')^2$, $D \equiv B'A - BA'$.
- $\mathcal{K}(v) \equiv \frac{v}{v^2}$; $X' \equiv \mathcal{K}(x'-w')$, $Y \equiv \mathcal{K}(y-z)$, $X_w \equiv \mathcal{K}(x-w)$, $X_z \equiv \mathcal{K}(x-z)$, $Z_w \equiv \mathcal{K}(z-w)$.
- The logarithms that carry a:

$$\lambda \equiv \log\Big[(1-a)\frac{A}{B} + a\Big], \qquad \lambda' \equiv \log\Big[(1-a)\frac{A'}{B'} + a\Big], \qquad \lambda_2 \equiv \log\Big|\frac{A}{B} - a\Big|.$$

- Brackets:

$$\mathbb{T}^{ij} \equiv \frac{\delta^{ij}}{2(z-w)^2} + \frac{(x-z)^j(z-w)^i}{(z-w)^2(x-z)^2} - \frac{(x-z)^j(x-w)^i}{(x-w)^2(x-z)^2} - \frac{(x-w)^i(z-w)^j}{(z-w)^2(x-w)^2},$$

$$\mathbb{L}^{ij} \equiv \frac{(x-w)^i(z-w)^j}{(z-w)^2(x-w)^2} + \frac{(x-z)^j(x-w)^i}{2(x-z)^2(x-w)^2} \quad \text{(the bracket of your } \log(1-\tfrac{k^+}{\vee})\text{ terms)}.$$

- Colour: $\mathcal{C}^a_L \equiv U^{ab'}(x')\rho^{b'}(x') - U^{ab'}(w')\rho^{b'}(x')$ and $\mathcal{R}^{ab}(w,z,x) \equiv f^{cbd}U^{ac}(w)\rho^d(x) - f^{acd}U^{bc}(z)U^{de}(x)\rho^e(x)$.
- LO and its fragmentation form, with $\mathcal{O} = [U^{e'c}(y') - U^{e'c}(x')][U^{ec}(y) - U^{ce}(x)]\rho^{e'}(x')\rho^e(x)$:

$$\mathrm{LO}\big[f\big] \equiv \frac{1}{(2\pi)^3}\frac{g^2}{2\pi^2 k^+}\int_{x,y,x',y'} e^{-ik\cdot(y'-y)}\, f\, \frac{(x-y)\cdot(x'-y')}{(x-y)^2(x'-y')^2}\,\mathcal{O},$$

$$\mathrm{LO}_\xi\big[f\big] \equiv \text{the same with } e^{-ik\cdot(y'-y)} \to e^{-ik\cdot(y'-y)\frac{1}{\xi}}, \qquad \mathrm{LO} \equiv \mathrm{LO}[1].$$

  In $\mathrm{LO}[\ell_X]$, $X = x - y$ (the emission distance); $\Delta \equiv y' - y$.
- $\mathcal{N} \equiv \frac{1}{(2\pi)^3}\frac{g^4}{16\pi^5}\frac{1}{k^+}$; $\mathcal{J}$ = JIMWLK⊗LO in units of $\mathcal{N}$ (`Final_evolution_check.pdf`).
- Special functions: $E_1(z) = \int_z^\infty \frac{e^{-t}}{t}dt$ (principal branch), $\mathrm{Ein}(z) = \int_0^z \frac{1-e^{-t}}{t}dt = E_1(z) + \log z + \gamma_E$, and

$$\mathcal{E}(\beta) \equiv \mathrm{Ein}(i\beta) + L_\vee = \gamma_E + \log\frac{i\beta}{a} + E_1(i\beta).$$

- $U^{bc} = U^{cb}$, as in your files.

4 Check of the file

Method (no limit in $\vee$, k^+ or Λ). For every row the first-principles integrand (`fp1.py`, `fixed23.py`) was integrated exactly over p^+ on its exact range: $\Lambda < p^+ < \vee - k^+$; $\Lambda < p^+ < k^+ - \Lambda$ and $k^+ + \Lambda < p^+ < \vee$ for Group I Row III (principal value); $\Lambda < p^+ < k^+ - \Lambda$ for Group I Row V. Then

$$D(\vee, \Lambda) = \text{exact} - \text{file}$$

was computed for $(\vee, \Lambda) = (6, 10^{-3}), (40, 10^{-3}), (6, 10^{-5}), (40, 10^{-5}), (300, 10^{-4})$, $k^+ = 1$. If the file is complete, $D/\log\frac{\vee}{\Lambda}$ is the same in all five cases.

Table 1: Check of `Finite_terms.tex` (three random configurations, SU(3)).

row	compared with	result (spread of $D/\log\frac{\vee}{\Lambda}$)
G1 Row II part 1, 2a, 2b	corrected Group I file	identical after removing $\log\frac{\vee}{\Lambda}$; typo F3; remark R
G1 Row III	exact principal value	7×10^{-13} (with $\log\lvert\cdot\rvert$, F1)
G1 Row IV	exact	2×10^{-17}
G1 Row V part 1	exact on $(\Lambda, k^+ - \Lambda)$	$9\times10^{-5} = O(\Lambda/k^+)$ (F2)
G1 Row V part 2	`Group_I_v2_check.pdf`	identical after removing $\log\frac{\vee}{\Lambda}$
G2 Row I parts 1, 2	exact	4×10^{-17}, 2×10^{-17}
G2 Row IV	`RowIV_FF_userform.pdf`	identical after removing $\log\frac{\vee}{\Lambda}$; F7
G3 Rows I, II	exact	3×10^{-17}, 3×10^{-17}
G3 Row III part 1, V, VI	exact	$1\times10^{-4} = O(\Lambda/k^+)$ (F2)
G3 Row III part 2	exact	$3\times10^{-5} = O(\Lambda/k^+)$ (F2); typo F4

Corrections.

F1 Group I Row III: the logarithms are principal values, $\log[\dots] \to \log|\dots|$ (pole at $p^+ = k^+B/A$).

F2 Plus prescriptions from 0 (Group I Row V; Group III Rows III, V, VI) are exact only with lower limit Λ. For $\Lambda \to 0$ this is harmless.

F3 Group I Row II, d^3N_{2a}: $\int_{x,y,x',z} \to \int_{x,y,x'}$.

F4 Group III Row III part 2: $\int_{w,w',y',z,y,x} \to \int_{w,x',y',z,x}$ (and $\int_{w,y',z,y,x} \to \int_{w,x',y',z,x}$ in the $\log(1-\frac{k^+}{\vee})$ term).

F5 Group I Row II part 1: $\int_{w,w',x,x',}$.

F6 **The four-ρ terms are not "NA".** With a exact they give a finite term of order 1 when $k^+ \sim \vee$: Sec. 9.

F7 Group II Row IV, third line: $-d_\perp 2\,\frac{(x-y)\cdot(y'-y)\,(x'-y')\cdot(y'-y)}{\dots}$ (no pole multiplies it).

R In `Group_I.tex` Row II 2a the f^{abc} line has the wrong sign; the finite file is correct.

5 $\Lambda \to 0$: the only limit taken

Exactly in a:

$$\log\frac{(\vee-k^+)A+k^+B}{\Lambda A+k^+B} \to L_\vee+\lambda, \qquad \log\left|\frac{\vee A-k^+B}{(\Lambda+k^+)A-k^+B}\right| \to L_\vee+\log\left|\frac{A-aB}{A-B}\right|,$$

$$\log\left|\frac{k^+B-\vee A}{k^+B-\Lambda A}\right| \to L_\vee+\lambda_2, \qquad \log\left|\frac{k^+B-\vee A}{k^+B-(k^++\Lambda)A}\right| \to L_\vee+\log\left|\frac{A-aB}{A-B}\right|,$$

$$\log\left|\frac{(k^+-\Lambda)A-k^+B}{\Lambda A-k^+B}\right| \to \log\left|\frac{A-B}{B}\right|, \qquad \log\frac{[k^+(x-w)-(k^+-\Lambda)(x-z)]^2}{[k^+(x-w)-\Lambda(x-z)]^2} \to \log\frac{(z-w)^2}{B},$$

$$\log\frac{\vee}{k^+\pm\Lambda},\ -\log\frac{k^+-\Lambda}{\vee} \to L_\vee, \qquad \frac{1}{\vee}-\frac{1}{k^++\Lambda} \to -\frac{1-a}{k^+},$$

$$\log\left(1-\frac{\Lambda}{k^+}\right) \to 0, \qquad \mathcal{A} \to \mathcal{A}_0,$$

with $\mathcal{A}_0 = \frac{1}{r}\big[\arctan\frac{A-(x-w)\cdot(x-z)}{r} + \arctan\frac{(x-w)\cdot(x-z)}{r}\big]$, $r = \sqrt{AB-[(x-w)\cdot(x-z)]^2}$.

- Limits of integration: $\frac{k^+}{k^++\Lambda} \to 1$, $k^+-\Lambda \to k^+$; plus prescriptions start at 0.
- The $\frac{1}{\Lambda}$ terms of Group I Row III cancel exactly (E3).
- Everything else, i.e. $\log(1-a)$, $\log^2(1-a)$, $L_\vee\log(1-a)$, the a inside R^i, the upper limits $\vee-k^+$ and $\frac{k^+}{\vee}$, is kept.

6 Running coupling

In units of LO (inside the integrals), both orderings (your c.c.), exactly in a:

$$\text{G1 Row II, part 1} = \frac{\alpha_s N_c}{4\pi}\,\text{LO}\Big[(\hat{E}_X+\hat{E}_{X'})\Big(\frac{11}{3}+2L_\vee+\log(1-a)\Big)\Big] + \frac{\alpha_s N_c}{2\pi}\Big[\frac{67}{9}-\frac{\pi^2}{3}-\log^2(1-a)-L_\vee\log(1-a)\Big]\text{LO}, \tag{6.1}$$

$$\text{G1 Row V, 2b UV} = \frac{\alpha_s N_c}{2\pi}\,\text{LO}\Big[(\hat{E}_X+\hat{E}_{X'})\Big(-2L_\vee-\frac{11}{6}\Big)\Big], \tag{6.2}$$

with $X = x-y$, $X' = x'-y'$ the emission distances of LO (Group I writes w, w' for y, y'; in 2b UV rename $y \to x$, $z \to y$). The bracket of Row II is $L_\vee+\log(\frac{\vee}{k^+}-1) = 2L_\vee+\log(1-a)$, exactly.

- **The β_0 terms cancel exactly** ($\frac{11}{3}$ against $2\times\frac{11}{6}$, with pole, γ_E, μ and $\log X^2$), for every a (`sympy`: difference 0).
- **Left over:**

$$\boxed{\left.\frac{d^3N}{d^3k}\right|_{\text{RC}} = \frac{\alpha_s}{2\pi}N_c\Big[\frac{67}{9}-\frac{\pi^2}{3}-\log^2\Big(1-\frac{k^+}{\vee}\Big)-\log\frac{\vee}{k^+}\log\Big(1-\frac{k^+}{\vee}\Big)\Big]\text{LO}} \tag{6.3}$$

- The terms $(\hat{E}_X+\hat{E}_{X'})(2L_\vee+\log(1-a))$ and $-4L_\vee(\hat{E}_X+\hat{E}_{X'})$ are not running coupling; they combine with Group II Row IV (Sec. 7).
- **The scale of α_s in LO.** The LO coupling is renormalized in $\overline{MS}$. Its counterterm is the $\frac{\beta_0}{2}\delta(1-\xi)$ part of the fragmentation counterterm (Sec. 7), so $\alpha_s = \alpha_s(\mu_F)$.

7 DGLAP, and the pole that is left over

Group II Row IV, lines 1–2. With $d_\perp = 2-\epsilon$ your weight is $W = -\hat{E}_\Delta + O(\epsilon)$. Your subtraction at $\xi = 1$ converts exactly (lower limit $\frac{k^+}{\vee}$ kept, `RowIV_FF_userform.pdf`, Eq. (3.3)):

$$\text{lines 1, 2} = \frac{\alpha_s}{2\pi}\int_a^1 \frac{d\xi}{\xi^2}\, P^{(r)}_{gg,+}(\xi)\,\mathrm{LO}_\xi\big[W\big] - \frac{\alpha_s}{2\pi}\,2N_c L_\vee\,\mathrm{LO}\big[W\big], \qquad P^{(r)}_{gg,+} = 2N_c\Big[\frac{\xi}{(1-\xi)_+} + \frac{1-\xi}{\xi} + \xi(1-\xi)\Big], \quad (7.1)$$

where $(1-\xi)_+$ is the standard distribution on $[0,1]$, restricted to $\xi > a$:

$$\int_a^1 d\xi\,\frac{f(\xi)}{(1-\xi)_+} = \int_a^1 d\xi\,\frac{f(\xi)-f(1)}{1-\xi} + f(1)\log(1-a).$$

Your second line ($\log(1-\frac{k^+}{\vee})$) is exactly this last term.

The poles, exactly in a. In units of $\frac{\alpha_s}{2\pi}(-\frac{2}{\epsilon})N_c\,\mathrm{LO}$ (the $\delta(1-\xi)$ terms):

source	coefficient
G2 Row IV, Eq. (7.1)	$-2L_\vee$
G1 Row II part 1, Eq. (6.1)	$-\frac{11}{3} - 2L_\vee - \log(1-a)$
G1 Row V 2b UV, Eq. (6.2)	$+\frac{11}{3} + 4L_\vee$
charge renormalization of LO	$+\frac{11}{6}$
sum	$\frac{11}{6} - \log(1-a)$ (`sympy`: $11/6 - \log(1-a)$)

So the full pole of the gluon spectrum is

$$\frac{d^3N_g}{d^3k}\bigg|_{\text{pole}} = \frac{\alpha_s}{2\pi}\Big(-\frac{2}{\epsilon}\Big)\Big[\int_{k^+/\vee}^1 \frac{d\xi}{\xi^2}\,P_{gg}(\xi)\,\mathrm{LO}_\xi \;-\; N_c\log\Big(1-\frac{k^+}{\vee}\Big)\mathrm{LO}\Big], \qquad P_{gg} = P^{(r)}_{gg,+} + \frac{11N_c}{6}\,\delta(1-\xi). \quad (7.2)$$

- **The first term** is absorbed in $D_{h/g}$, exactly. The hadron convolution has the lower limit $\frac{k^+}{\vee}$ (the gluon momentum $\frac{k^+}{\zeta}$ must be below $\vee$), and the counterterm $D^{(0)} = D + \frac{\alpha_s}{2\pi}\frac{2}{\epsilon}P_{gg}\otimes D$ cancels it for every a (`RowIV_FF_userform.pdf`, check $B = -A$ with the cut supports).
- **The second term** is new and is not absorbed:

$$\boxed{\frac{d^3N}{d^3k}\bigg|_{\text{rem}} = \frac{\alpha_s N_c}{4\pi}\,\log\Big(1-\frac{k^+}{\vee}\Big)\,\mathrm{LO}\big[\hat{E}_X + \hat{E}_{X'}\big] \;\ni\; \frac{\alpha_s}{2\pi}\,\frac{2}{\epsilon}\,N_c\log\Big(1-\frac{k^+}{\vee}\Big)\mathrm{LO}} \quad (7.3)$$

  - Its source is the $\log(\frac{\vee}{k^+}-1)$ of Group I Row II part 1, i.e. the limit $\xi \geq \frac{k^+}{\vee}$ of the dressed one-ρ state $|\Psi^B_{g\rho}\rangle$ (correction X6).
  - In the real rows the same limit appears only inside the standard plus distribution, so it is part of $P_{gg}\otimes\mathrm{LO}$.
  - It depends on $\frac{k^+}{\vee}$, not on ξ alone, so it cannot be part of $D_{h/g}$ or of α_s.
  - It vanishes only for $k^+ \ll \vee$.
  - In AABKL the virtual term is $\int_0^1 d\xi$ with no limit from the projectile momentum, and only the real terms carry $\int_0^{1-x_p}$ (Sec. 13). See Q1.

The finite parts. `sympy` (difference 0):

$$\boxed{\frac{d^3N}{d^3k}\bigg|_{L_\vee\hat{E}} = \frac{\alpha_s N_c}{2\pi}\,\log\frac{\vee}{k^+}\,\mathrm{LO}\Big[\log\frac{(y-y')^4}{(x-y)^2(x'-y')^2}\Big] \quad \text{(pole and } \mu \text{ cancel)},} \quad (7.4)$$

$$\boxed{\frac{d^3N}{d^3k}\bigg|_{\text{DGLAP}} = -\frac{\alpha_s}{2\pi}\int_{k^+/\vee}^1 \frac{d\xi}{\xi^2}\,P^{(r)}_{gg,+}(\xi)\,\mathrm{LO}_\xi\big[\ell_\Delta(\mu_F)\big], \qquad \ell_\Delta = \log\frac{(y'-y)^2\mu_F^2}{c_0^2}} \quad (7.5)$$

- The lower limit $\frac{k^+}{\vee}$ is kept; the previous "reg" is withdrawn.
- μ_F dependence: $\mu_F^2\frac{\partial}{\partial\mu_F^2}D_{h/g} = \frac{\alpha_s}{2\pi}P_{gg}\otimes D_{h/g}$ (DGLAP), with the $\frac{\beta_0}{2}\delta(1-\xi)$ from the running of $\alpha_s(\mu_F)$ in LO.
- With $d_\perp = 2$: $\ell_\Delta \to \ell_\Delta + 1$.

Line 3 (linear polarization) , with $\Lambda \to 0$ only in the upper limit:

$$\frac{d^3N_{\rm lin}}{d^3k} = \frac{\alpha_s}{2\pi}\int_{k^+/\vee}^1 \frac{d\xi}{\xi^2}\, 2N_c\,\xi(1-\xi)\,\frac{1}{(2\pi)^3}\frac{g^2}{2\pi^2 k^+}\int_{x,y,x',y'} e^{-ik\cdot(y'-y)\frac{1}{\xi}}\Big[X\cdot X' - 2(X\cdot\hat\Delta)(X'\cdot\hat\Delta)\Big]\mathcal{O}, \tag{7.6}$$

with $X = \mathcal{K}(x-y)$, $X' = \mathcal{K}(x'-y')$, $\hat\Delta = \frac{y'-y}{|y'-y|}$. The tensor is traceless.

8 The remaining finite terms, exact in $\vee$

Each row keeps its prefactor, $\frac{1}{(2\pi)^3}\int e^{-ik\cdot(w'-w)}$ (or its own phase) and its colour factor, as in the file. $p_8 \equiv \frac{ig^4}{8\pi^5 k^+}$, $p_{16} \equiv \frac{ig^4}{16\pi^5 k^+}$. Checks against your expressions: Sec. 15.

8.1 Group I

Row II. Part 1: Secs. 6, 7. Part 2a: unchanged (three ρ; $L_\vee$ exact). Part 2b: unchanged, with the exact

$$R^i = -2k^i\log\frac{p^2}{k^2+a\,p^2} + \frac{(2p^2-2k\cdot p+k^2)k^i + k^2p^i}{(k-p)^2}\log\Big[\frac{k^2}{(k-p)^2} + a\Big(1-\frac{k^2}{(k-p)^2}\Big)\Big] - \frac{p^2k^i+k^2p^i}{(k-p)^2}\log(1-a)$$

(the previous R^i_∞ is withdrawn).

Row III (all of N_1, N_2, N_{3a}, N_{3b}), colour $\mathcal{C}^{\rm III} = f^{abd}\mathcal{C}^a_L[U^{de}(x)U^{bc}(z)\rho^e(x)\rho^c(y) - U^{de}(x)U^{bc}(y)\rho^e(x)\rho^c(y)]$:

$$\begin{aligned}\frac{d^3N_{\rm III}}{d^3k} = \frac{1}{(2\pi)^3}\int_{w,w'} e^{-ik\cdot(w'-w)}\int_{x,y,z,x'}\Big\{&p_8L_\vee\Big[(X'\cdot X_w)(Y\cdot Z_w - Y\cdot X_z) + \frac{X'\cdot Y}{(z-w)^2} - \Sigma_4\Big]\\ &+\lambda_2\Big[p_{16}T_1 - p_8(X'\cdot X_w)(X_z\cdot Y) + p_8(Q-\Sigma_4)\Big]\\ &-\log(1-a)\,[p_{16}T_1 + p_8Q] + p_8\mathcal{A}_0M_0 + N_{3b}\Big\}\,\mathcal{C}^{\rm III},\end{aligned} \tag{8.1}$$

$$\begin{aligned}T_1 &= (X'\cdot Y)\frac{A+B}{(A-B)^2}, \qquad Q = (X'\cdot Y)\,\frac{[(z-w)\cdot(x-z)]B - [(z-w)\cdot(x-w)]A}{(z-w)^2(B-A)^2},\\ \Sigma_4 &= \frac{(X'\cdot Z_w)\,Y\cdot(x-w) - (X'\cdot X_w)(Y\cdot Z_w)A - (X'\cdot Z_w)(Y\cdot X_z)B + X'\cdot(x-z)\,(Y\cdot Z_w)}{B-A},\\ M_0 &= (X'\cdot Z_w)\,Y\cdot(x-w) + (Y\cdot Z_w)\,X'\cdot(x-z) - \frac{(x-w)\cdot(x-z)}{A}\,(X'\cdot Z_w)\,Y\cdot(x-z),\\ N_{3b} &= -p_8\mathcal{A}_0\frac{(x-w)\cdot(x-z)}{B}(Y\cdot Z_w)\,X'\cdot(x-w)\\ &\quad + p_{16}\log\frac{(z-w)^2}{B}\Big[-\frac{(X'\cdot Z_w)\,Y\cdot(x-z)}{A} + \frac{(Y\cdot Z_w)\,X'\cdot(x-w)}{B} - \frac{X'\cdot Y}{(z-w)^2}\Big].\end{aligned}$$

- The a dependence sits only in $\lambda_2 = \log|\frac{A}{B} - a|$, $\log(1-a)$ and $L_\vee$.
- The six logarithms of N_1, N_2, N_{3a} combine as $\log|\frac{A-aB}{(1-a)B}| = \lambda_2 - \log(1-a)$ and $\log|\frac{aB-A}{a(B-A)}| + \log|\frac{B-A}{B}| = L_\vee + \lambda_2$.

Row IV.

$$\begin{aligned}\frac{d^3N_{\rm IV}}{d^3k} &= \frac{1}{(2\pi)^3}\int e^{-ik\cdot(w'-w)}\,p_8\,\mathbb{K}_{\rm IV}\times\text{colour of Row IV},\\ \mathbb{K}_{\rm IV} &\equiv X'^iY^j\mathbb{T}^{ij}\big(L_\vee+\lambda\big) - \frac{X'\cdot Y}{2(z-w)^2}L_\vee + \log(1-a)\,X'^iY^j\mathbb{L}^{ij}.\end{aligned} \tag{8.2}$$

Row V, part 1. The p^+ and q^+ integrals on $(0,k^+)$ in closed form, $\beta \equiv k\cdot(z-x)$:

$$\begin{aligned}\frac{d^3N_{\rm V1}}{d^3k} = \frac{1}{(2\pi)^3}\frac{ig^4}{32\pi^5}\frac{1}{k^+}\int_{x,z,y,x',y',w'}&\Big\{e^{-ik\cdot(w'-z)}\Big[\kappa_a\,\frac{1-e^{-i\beta}}{i\beta} + \kappa_b\,\mathcal{E}(\beta)\Big] + e^{-ik\cdot(w'-x)}\,\kappa_q\,\mathcal{E}(-\beta)\Big\}\\ &\times\Big[f^{adc}U^{db}(x)U^{ce}(z)\{\rho^e(y),\rho^b(y')\} - f^{abc}U^{cd}(y)U^{be}(y')\{\rho^d(y),\rho^e(y')\}\Big]\mathcal{C}^a_L,\end{aligned} \tag{8.3}$$

$\kappa_a = [X'\cdot\mathcal{K}(z-x)][Y\cdot\mathcal{K}(y'-x)]$, $\kappa_b = [\mathcal{K}(y'-x)\cdot\mathcal{K}(z-x)](X'\cdot Y)$, $\kappa_q = [X'\cdot\mathcal{K}(y'-x)][\mathcal{K}(z-x)\cdot Y]$. The $L_\vee$ of your $\log\frac{k^+-\Lambda}{\vee}$ terms is inside $\mathcal{E}$.

Row V, part 2 (2a, 2b finite). With $t = p^+/k^+$ (or q^+/k^+), $Y_x \equiv (y-x)^2$, $Y_z \equiv (y-z)^2$, $\beta \equiv k\cdot(z-x)$, define the master integrals

$$\mathcal{M}_n \equiv \int_0^1 dt\, \frac{t^n\, e^{-i\beta t}}{tY_x + (1-t)Y_z}, \qquad \mathcal{N}_0 \equiv \int_0^1 dt\, \frac{e^{i\beta t}}{(1-t)Y_x + tY_z}.$$

In closed form, with $\delta \equiv Y_x - Y_z$, $c \equiv Y_z/\delta$, $m_0 = \frac{1-e^{-i\beta}}{i\beta}$, $m_1 = \frac{m_0 - e^{-i\beta}}{i\beta}$:

$$\mathcal{M}_0 = \frac{e^{i\beta c}}{\delta}\big[E_1(i\beta c) - E_1(i\beta(1+c))\big], \quad \mathcal{M}_1 = \frac{m_0}{\delta} - c\,\mathcal{M}_0, \quad \mathcal{M}_2 = \frac{m_1 - c\,m_0}{\delta} + c^2\mathcal{M}_0,$$

and $\mathcal{N}_0$ is $\mathcal{M}_0$ with $\beta \to -\beta$, $Y_x \leftrightarrow Y_z$. Contract your brackets with $X'^i\mathcal{K}^m(z-x)$:

$$\begin{aligned}
s_1 &= X'\cdot\mathcal{K}(z-x)\,\frac{d_\perp}{2(x-z)^2}(Y_x - Y_z),\\
v_2 &= X'\cdot\mathcal{K}(z-x)\Big[\frac{(y-z)\cdot(x-z)}{(x-z)^2} + \frac{(y-x)\cdot(y-z)}{2Y_x} - \frac{Y_x - Y_z}{2(x-z)^2}\Big],\\
v_4 &= X'\cdot\mathcal{K}(z-x)\Big[\frac{(y-x)\cdot(x-z)}{(x-z)^2} - \frac{(y-x)\cdot(y-z)}{2Y_z} - \frac{Y_x - Y_z}{2(x-z)^2}\Big],\\
s_3 &= \frac{X'\cdot(y-z)\,\mathcal{K}(z-x)\cdot(x-z)}{(x-z)^2} + \frac{\mathcal{K}(z-x)\cdot(y-x)\,X'\cdot(y-z)}{2Y_x},\\
s_5 &= \frac{\mathcal{K}(z-x)\cdot(y-x)\,X'\cdot(x-z)}{(x-z)^2} - \frac{\mathcal{K}(z-x)\cdot(y-x)\,X'\cdot(y-z)}{2Y_z}\\
&\quad + \frac{\mathcal{K}(z-x)\cdot(y-z)\,X'\cdot(x-z)}{(x-z)^2} + \frac{X'\cdot(y-x)\,\mathcal{K}(z-x)\cdot(y-z)}{2Y_x},\\
s_6 &= \frac{X'\cdot(y-x)\,\mathcal{K}(z-x)\cdot(x-z)}{(x-z)^2} - \frac{X'\cdot(y-x)\,\mathcal{K}(z-x)\cdot(y-z)}{2Y_z}.
\end{aligned}$$

Then, exactly (all p^+, q^+ integrals, plus prescriptions and $\log\frac{k^+-\Lambda}{\vee}$ terms included; $d_\perp = 2$):

$$\begin{aligned}
\{\dots\}_{\mathrm{V2}} \equiv\; & e^{-ik\cdot(w'-z)}\Big[s_1(\mathcal{M}_1 - \mathcal{M}_2) + v_2(\mathcal{M}_0 - \mathcal{M}_1) + v_4\mathcal{M}_1 + \Big(s_3\frac{Y_x}{Y_z} - s_5\Big)\mathcal{M}_0 + \frac{s_3}{Y_z}\mathcal{E}(\beta)\Big]\\
& + e^{-ik\cdot(w'-x)}\,s_6\Big[\frac{Y_z}{Y_x}\mathcal{N}_0 + \frac{1}{Y_x}\mathcal{E}(-\beta)\Big],
\end{aligned} \tag{8.4}$$

$$\frac{d^3N_{2a}}{d^3k} = \frac{1}{(2\pi)^3}\frac{g^4}{16\pi^5}\frac{1}{k^+}\int_{x,y,z,x',w'} \{\dots\}_{\mathrm{V2}}\,\big[U(x') - U(w')\big]^{ab'}\big[-f^{adc}f^{fbe}U^{db}(x)U^{ce}(z) + N_cU^{af}(x)\big]\rho^{b'}(x')\rho^f(y), \tag{8.5}$$

$$\begin{aligned}
\frac{d^3N_{2b,\mathrm{fin}}}{d^3k} = \frac{1}{(2\pi)^3}\frac{g^4}{16\pi^5}\frac{1}{k^+}\int_{x,y,z,x',w'} & \Big\{\{\dots\}_{\mathrm{V2}}\,\big[U(x') - U(w')\big]N_c\big[U(y) - U(x)\big]\\
& + e^{-ik\cdot(w'-z)}\Big(\frac{11}{6} + 2L_\vee\Big)\frac{X'\cdot Y\;(y-x)\cdot(y-z)}{(y-x)^2(x-z)^2}\,\big[U(x') - U(w')\big]N_c\big[U(y) - U(z)\big]\Big\}
\end{aligned} \tag{8.6}$$

In Eq. (8.6) the colour factors are $[U^{ab'}(x') - U^{ab'}(w')]N_c[U^{af}(y) - U^{af}(x)]\rho^{b'}(x')\rho^f(y)$ and the same with $U^{af}(x) \to U^{af}(z)$, as in the file. Part 2b UV: Sec. 6.

8.2 Group II

Row I, part 1 (three ρ). Block A: $-p_{16}\,\mathbb{K}_{\mathrm{IV}}$; block B: $+p_{16}\,\mathbb{K}_{\mathrm{IV}}$ with $w \leftrightarrow w'$ in the kernel and $X' \to \mathcal{K}(x'-w)$. Colours as in the file.

Row I, part 2 (2a–2f, two ρ). Common factor $\frac{1}{(2\pi)^3}\frac{g^4}{8\pi^5}\frac{1}{k^+}\int e^{-ik\cdot(w'-w)}\,\mathcal{C}_{58}$, with $\mathcal{C}_{58}$ the colour of the file and $[\dots]_{2a}$–$[\dots]_{2e}$ its brackets:

$$\begin{aligned}
&2a:\ \frac{[\dots]_{2a}}{D}(\lambda - \lambda'), \qquad 2b:\ [\dots]_{2b}\Big(\frac{\log(1-a) - L_\vee}{BB'} - \frac{A\lambda}{BD} + \frac{A'\lambda'}{B'D}\Big), \qquad 2c:\ [\dots]_{2c}\Big(\frac{L_\vee}{AA'} + \frac{\frac{B'}{A'}\lambda' - \frac{B}{A}\lambda}{D}\Big),\\
&2d:\ [\dots]_{2d}\Big(\frac{A\lambda}{(A-B)D} + \frac{A'\lambda'}{(B'-A')D}\Big), \qquad 2e:\ -[\dots]_{2e}\Big(\frac{B\lambda}{(A-B)D} + \frac{B'\lambda'}{(B'-A')D}\Big),\\
&2f:\ \frac{d_\perp}{4}\,\frac{1}{(z-w)^2(z-w')^2}\Big[-\frac{AB(A'-B')}{(A-B)D}\lambda + \frac{A'B'(A-B)}{(A'-B')D}\lambda' - (1-a)\Big].
\end{aligned}$$

- In 2d, 2e, 2f the explicit $\log\frac{\vee}{k^+ + \Lambda}$ cancels exactly against the $L_\vee$ inside the other logarithms (`sympy`).
- In 2b the $\log(1-a)$ term of the file simplifies to $\frac{\log(1-a)}{BB'}$.

Rows II, III. Four ρ only: Sec. 9.

Row IV. Sec. 7: Eqs. (7.1), (7.5), (7.6).

8.3 Group III

Rows I, II. $-p_8\,\mathbb{K}_{\rm IV}$ with the colour of the row.

Row III part 1, Rows V, VI. With $\kappa \equiv k\cdot(y'-z)$ the p^+ integral (with the plus prescription and your $\log(1-\frac{k^+}{\vee})$ term) is

$$\frac{d^3N_C}{d^3k} = \frac{1}{(2\pi)^3}\frac{c_C}{k^+}\int_{w,x',y',z,y,x} e^{-ik\cdot(y'-w)}\Big\{t_1\,e^{i\kappa}\Big[E_1(i\kappa)-E_1\Big(\frac{i\kappa}{a}\Big)\Big] - t_3\,\frac{1-e^{-i\kappa\frac{1-a}{a}}}{i\kappa} + t_2\Big[\gamma_E+\log\frac{i\kappa}{a}+E_1\Big(i\kappa\frac{1-a}{a}\Big)\Big]\Big\}\times\text{colour}, \tag{8.7}$$

with $c_{\rm III(1)} = \frac{ig^4}{16\pi^5}$, $c_{\rm V} = c_{\rm VI} = \frac{ig^4}{8\pi^5}$, the colour factors of the file (with their $f^{c'd'a}$, $f^{c'ca}$), and

$$t_1 = [\mathcal{K}(y'-z)\cdot\mathcal{K}(x'-y')][\mathcal{K}(y-w)\cdot\mathcal{K}(x-z)], \qquad t_2 = [\mathcal{K}(y'-z)\cdot\mathcal{K}(x-z)][\mathcal{K}(y-w)\cdot\mathcal{K}(x'-y')],$$
$$t_3 = [\mathcal{K}(y'-z)\cdot\mathcal{K}(y-w)][\mathcal{K}(x-z)\cdot\mathcal{K}(x'-y')].$$

- As $y'\to z$ ($\kappa\to 0$) the brace stays finite: $t_1 L_\vee - t_3\frac{1-a}{a} - t_2\log(1-a)$. So the y' integral converges for every a.
- The previous caution, which concerned $\vee\to\infty$ taken pointwise in y', does not apply here.

Row III part 2. With $b\equiv B/A$, $U_a\equiv\frac{1-a}{a}$, $\mathcal{J}(b)\equiv e^{i\kappa b}[E_1(i\kappa b)-E_1(i\kappa(b+U_a))]$, $\mathcal{J}_0\equiv\frac{1-e^{-i\kappa U_a}}{i\kappa}$, $\mathcal{G}\equiv\gamma_E+\log\frac{i\kappa}{a}+E_1(i\kappa U_a)$, the five p^+ integrals ($D_p = p^+A+k^+B$, $P=p^++k^+$) are

$$\begin{aligned}&\int_0^{\vee-k^+} dp^+ e^{-i\kappa\frac{p^+}{k^+}}\Big\{\frac{p^+}{P^2D_p},\frac{p^+}{PD_p},\frac{1}{PD_p},\frac{p^+}{D_p},\frac{1}{D_p}\Big\}\\ &\equiv\{\mathbb{I}_1,\dots,\mathbb{I}_5\}\\ &=\Big\{\frac{1}{Ak^+}\Big[\frac{b\,(\mathcal{J}(1)-\mathcal{J}(b))}{(1-b)^2}+\frac{1-a\,e^{-i\kappa U_a}-i\kappa\mathcal{J}(1)}{1-b}\Big],\ \frac{\mathcal{J}(1)-b\,\mathcal{J}(b)}{A(1-b)},\\ &\quad\frac{\mathcal{J}(1)-\mathcal{J}(b)}{k^+A(b-1)},\ \frac{k^+}{A}\big(\mathcal{J}_0-b\,\mathcal{J}(b)\big),\ \frac{\mathcal{J}(b)}{A}\Big\}.\end{aligned}$$

Writing your bracket as $\frac{1}{D_p}\big[c_1\frac{p^+}{P^2}+c_2\frac{p^+}{k^+P}+c_3\frac{1}{P}+c_4\frac{p^+}{k^{+2}}+c_5\frac{1}{k^+}\big]+$ (the $[\frac{1}{p^+}]_+$ term with coefficient s), where

- $u\equiv\mathcal{K}(y'-z)$, $v\equiv\mathcal{K}(x'-y')$;
- $c_1 = (u\cdot v)\frac{d_\perp(A-B)}{2(z-w)^2}$; c_2, $c_3 = (u\cdot v)\times$ your brackets of $\frac{p^+}{k^+(p^++k^+)}$ and $\frac{1}{p^++k^+}$;
- $c_4 = -\frac{(v\cdot(x-z))(u\cdot(z-w))}{(z-w)^2}+\frac{(v\cdot(x-z))(u\cdot(x-w))}{2B}$;
- $c_5 = -[\dots]$ of your $\frac{1}{k^+}$ line, contracted with $u^m v^{k'}$;
- $s = -\frac{(v\cdot(x-w))(u\cdot(z-w))}{(z-w)^2}-\frac{(u\cdot(x-z))(v\cdot(x-w))}{2A}$;

the row is ($\mathcal{C}^{\rm III,2}$ = the colour factor of the file)

$$\frac{d^3N_{\rm III(2)}}{d^3k} = \frac{1}{(2\pi)^3}\frac{g^4}{8\pi^5}\int_{w,x',y',z,x} e^{-ik\cdot(y'-w)}\Big\{c_1\mathbb{I}_1+\frac{c_2}{k^+}\mathbb{I}_2+c_3\mathbb{I}_3+\frac{c_4}{k^{+2}}\mathbb{I}_4+\frac{c_5}{k^+}\mathbb{I}_5-\frac{s}{k^+B}\big[\mathcal{G}+\mathcal{J}(b)\big]\Big\}\mathcal{C}^{\rm III,2}. \tag{8.8}$$

9 The four-ρ terms with exact $\vee$ (F6)

Ranges. The four-ρ p^+ integrands are exactly $\propto\frac{1}{p^+}$. The lines of Eq. (4rho) of `4rho.tex` come from:

- Group I Row I (two lines) and Group I Row III, with range $(\Lambda,\vee)$: $\log\frac{\vee}{\Lambda}$;
- Group I Row IV, Group II Rows I, II, III and Group III Rows I, II, IV, with range $(\Lambda,\vee-k^+)$: $\log\frac{\vee}{\Lambda}+\log(1-a)$.

The $\log\frac{\vee}{\Lambda}$ parts cancel after symmetrization (max. $|\Sigma|$ 3×10^{-14} (terms up to 221)). The $\log(1-a)$ parts of the second set alone do *not*: max. $|\Sigma| = 12.3$–24.3 (terms up to 163). Hence, exactly, the finite four-ρ term is

$$\begin{aligned}\frac{d^3N_{4\rho}}{d^3k} &= \mathcal{N}\,\log\Big(1-\frac{k^+}{\vee}\Big)\int_{x,y,z,u,x',w,w'} e^{-ik\cdot(w'-w)}\,\mathcal{K}(x'-w')\cdot\mathcal{K}(u-w)\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)\\ &\quad\times\Big\{-\mathcal{C}_L^a\mathfrak{A}_3^a+\mathfrak{B}^{ab}\mathfrak{A}_B^{ab}+2\mathfrak{X}_1^{ab}\mathfrak{A}_1^{ab}+2\mathfrak{X}_2^{ab}\mathfrak{A}_2^{ab}+\big(\mathfrak{X}_1^{ab}+\mathfrak{X}_2^{ab}\big)\mathfrak{A}_B^{ab}+2\mathfrak{X}_2^{ab}\mathfrak{A}_1^{ab}\Big\}\\ &\quad+\text{c.c. of the G1 and G3 terms,}\end{aligned}\tag{9.1}$$

the six terms being G1 IV, G2 I, G2 II, G2 III, G3 I+II and G3 IV. Here, as in `4rho.tex`:

$$\begin{aligned}\mathfrak{A}_3^a &= U^{ac}(w)U^{bd}(z)U^{be}(y)\rho^e(y)\{\rho^d(x),\rho^c(u)\}-U^{bc}(y)\rho^c(y)\big(U^{bd}(x)\rho^d(x)U^{ae}(u)\rho^e(u)+U^{ae}(u)\rho^e(u)U^{bd}(x)\rho^d(x)\big),\\ \mathfrak{B}^{ab} &= \tfrac12\Big[\{\rho^b(x),\rho^{c'}(x')\}U^{c'a}(w')-\big(U^{c'd'}(x)\rho^{d'}(x)U^{ad}(x')\rho^d(x')+U^{ad}(x')\rho^d(x')U^{c'd'}(x)\rho^{d'}(x)\big)U^{bc'}(z)\Big],\\ \mathfrak{X}_1^{ab} &= U^{ab'}(x')\rho^{b'}(x')U^{c'a'}(x)\rho^{a'}(x)U^{c'b}(z)-\rho^{c'}(x')U^{bd'}(z)U^{d'a'}(x)\rho^{a'}(x)U^{c'a}(w'),\\ \mathfrak{X}_2^{ab} &= U^{c'd'}(x)\rho^{d'}(x)U^{ae'}(x')\rho^{e'}(x')U^{bc'}(z)-\rho^b(x)U^{ab'}(x')\rho^{b'}(x'),\\ \mathfrak{A}_B^{ab} &= U^{ac}(w)\{\rho^c(u),\rho^b(y)\}-U^{bc}(z)\big(U^{ce}(y)\rho^e(y)U^{ab'}(u)\rho^{b'}(u)+U^{ab'}(u)\rho^{b'}(u)U^{ce}(y)\rho^e(y)\big),\\ \mathfrak{A}_1^{ab} &= U^{bc}(z)U^{cd}(y)\rho^d(y)U^{ae}(u)\rho^e(u)-U^{ac}(w)U^{de}(y)\rho^e(y)U^{db}(z)\rho^c(u),\\ \mathfrak{A}_2^{ab} &= U^{bc}(z)U^{ad}(u)\rho^d(u)U^{ce}(y)\rho^e(y)-U^{ac}(u)\rho^c(u)\rho^b(y).\end{aligned}$$

- The operator products are in the order of `4rho.tex`. Their reordering into symmetric four-ρ, three-ρ and two-ρ parts is that of `4rho.tex`, restricted to these lines.
- If, as your kinematics $\Lambda < p^+ < \vee$ says, the independently emitted gluon has $p^+ < \vee$ in *every* row (also in Groups II and III), then all ranges are $(\Lambda,\vee)$ and Eq. (9.1) is zero. See Q2.

10 Three-ρ terms

These relations are exact for every a, because the rows involved share the range $(\Lambda,\vee-k^+)$:

1. **Group I Row IV + Group III Row I** $=0$ at the level of the p^+ integrand, for every p^+ and configuration (symmetrized charges; max. rel. 6×10^{-15}).
2. **Group II Row I part 1 (A) + Group III Row II** (rel. 4×10^{-16}):

$$\frac{1}{(2\pi)^3}\int e^{-ik\cdot(w'-w)}\int_{x,y,z,x'} p_8\,\mathbb{K}_{\rm IV}\,\mathcal{C}_L^a\rho^b(y)\mathcal{R}^{ab}(w,z,x), \qquad \text{block B and the c.c. of Row II: its mirror image.}\tag{10.1}$$

3. **The rest does not cancel.** The network $f^{abd}\mathcal{C}_L^a(U(z)\rho(y))^b(U(x)\rho(x))^d$ of Group I Row III, Eq. (8.1), against the same network in Eq. (10.1) differs by 0.16–8.6 (relative) at $\vee/k^+ = 1.5, 3, 30$. The reason is the energy denominators: p^+A-k^+B (virtual) against p^+A+k^+B (real).

- These symmetrized three-ρ finite terms are the classical first saturation correction of the projectile.
- They vanish for a Gaussian (MV) projectile (Q4).
- Their commutator (two-ρ) parts follow from the reordering identity $ABC = S(ABC)+\frac14(\{[A,B],C\}+\{[A,C],B\}+\{[B,C],A\})$.
- Row by row, they are the three-ρ $\to$two-ρ list of `Evolution_terms.tex`, with $\log\frac{\vee}{\Lambda}\times$kernel replaced by the exact kernel of Sec. 8.

11 Final result (exact in $\vee$)

For hadron production ($\overline{MS}$, $d_\perp = 2-\epsilon$, $n_f = 0$, your rule A; $a = \frac{k^+}{\vee}$, $L_\vee = \log\frac{\vee}{k^+}$):

$$
\begin{aligned}
\frac{d^3N_h}{d^3k} &= \int_{k^+/\vee}^1 \frac{d\zeta}{\zeta^3}\, D_{h/g}(\zeta,\mu_F)\, \frac{d^3N_g}{d^3k}\Big(\frac{k^+}{\zeta}, \frac{\mathbf{k}}{\zeta}\Big),\\
\frac{d^3N_g}{d^3k} &= \mathrm{LO}\big|_{\alpha_s(\mu_F)} + \mathcal{N}\log\frac{\vee}{\Lambda}\,\mathcal{J}\\
&\quad + \frac{\alpha_s}{2\pi}N_c\Big[\frac{67}{9} - \frac{\pi^2}{3} - \log^2(1-a) - L_\vee\log(1-a)\Big]\mathrm{LO} + \frac{\alpha_s N_c}{2\pi}L_\vee\,\mathrm{LO}\Big[\log\frac{(y-y')^4}{(x-y)^2(x'-y')^2}\Big]\\
&\quad - \frac{\alpha_s}{2\pi}\int_a^1 \frac{d\xi}{\xi^2}P^{(r)}_{gg,+}(\xi)\,\mathrm{LO}_\xi\big[\ell_\Delta(\mu_F)\big] + \frac{d^3N_{\mathrm{lin}}}{d^3k} + \frac{\alpha_s N_c}{4\pi}\log(1-a)\,\mathrm{LO}\big[\hat{E}_X + \hat{E}_{X'}\big]\\
&\quad + \Big[\text{G1 II 2b} + \text{G1 V 2a} + \text{G1 V 2b fin} + \text{G2 I(2)} + \text{G3 III(2)}\Big]_{\text{two }\rho}\\
&\quad + \Big[\text{G1 II 2a} + \text{G1 III} + \text{G1 V(1)} + \text{Eq. (10.1)} + \text{mirror} + \text{G3 III(1)} + \text{G3 V} + \text{G3 VI}\Big]_{\text{three }\rho}\\
&\quad + \frac{d^3N_{4\rho}}{d^3k} + O(\alpha_s^3) + O\Big(\frac{\Lambda}{k^+}\Big).
\end{aligned}
\tag{11.1}
$$

- The terms are: LO; your evolution terms; Eq. (6.3); Eq. (7.4); Eq. (7.5); Eq. (7.6); Eq. (7.3); Sec. 8; Secs. 8 and 10; Eq. (9.1).
- LO is $O(\alpha_s)$; every other term is $O(\alpha_s^2)$.
- Nothing is expanded in $\frac{k^+}{\vee}$. Λ appears only in $\log\frac{\vee}{\Lambda}$.
- **Two terms need a decision** (Q1, Q2):
  - the remainder $\frac{\alpha_s N_c}{4\pi}\log(1-a)\mathrm{LO}[\hat{E}_X + \hat{E}_{X'}]$ still contains a pole;
  - $d^3N_{4\rho}$ is zero if the independently emitted gluon has $p^+ < \vee$ in all rows.
- The lower limit $\frac{k^+}{\vee}$ of the ζ integral is the analogue of AABKL's x_F.

12 The final result of AABKL (arXiv:1411.2869)

T. Altinoluk, N. Armesto, G. Beuf, A. Kovner, M. Lublinsky, *Single-inclusive particle production in proton-nucleus collisions at next-to-leading order in the hybrid formalism.* Their final result ("This is the final result of this paper") is Eq. (4.20) with Eqs. (4.21) (quarks) and (4.22) (gluons):

$$
\begin{aligned}
\frac{d\sigma^H}{d^2p_{h\perp}d\eta} &= \frac{1}{(2\pi)^2}\int_{x_F}^1\frac{d\zeta}{\zeta^2}D^q_{H,\mu^2}(\zeta)\frac{x_F}{\zeta}f^q_{\mu^2}\Big(\frac{x_F}{\zeta}\Big)\int_{y\bar{y}} e^{i\frac{p_{h\perp}}{\zeta}(y-\bar{y})}s[y,\bar{y}] + \int_{x_F}^1\frac{d\zeta}{\zeta^2}D^q_{H,\mu^2}(\zeta)\frac{d\bar{\sigma}^q}{d^2p_\perp d\eta}\Big(\frac{p_{h\perp}}{\zeta},\frac{x_F}{\zeta}\Big)\\
&\quad + \frac{1}{(2\pi)^2}\int_{x_F}^1\frac{d\zeta}{\zeta^2}D^g_{H,\mu^2}(\zeta)\frac{x_F}{\zeta}f^g_{\mu^2}\Big(\frac{x_F}{\zeta}\Big)\int_{y\bar{y}} e^{i\frac{p_{h\perp}}{\zeta}(y-\bar{y})}s_A[y,\bar{y}] + \int_{x_F}^1\frac{d\zeta}{\zeta^2}D^g_{H,\mu^2}(\zeta)\frac{d\bar{\sigma}^g}{d^2p_\perp d\eta}\Big(\frac{p_{h\perp}}{\zeta},\frac{x_F}{\zeta}\Big)\\
&\quad + (q\to\bar{q};\ p_{h\perp}\to -p_{h\perp}),
\end{aligned}
\tag{AABKL 4.20}
$$

$$
\begin{aligned}
\frac{d\bar{\sigma}^g_1}{d^2p_\perp d\eta} &= \frac{g^2}{(2\pi)^3}\int_{y\bar{y}z} e^{ip_\perp(y-\bar{y})}\Bigg\{\int_0^{1-x_p}d\xi\,\frac{x_p}{1-\xi}f^g_{\mu^2}\Big(\frac{x_p}{1-\xi}\Big)\Big[\Big(\frac{1-\xi}{[\xi]_+} + \frac{\xi}{1-\xi} + \xi(1-\xi)\Big)\\
&\quad\times\Big(2C_A\Big[A^i(y-z)A^i(\bar{y}-z) - \big(A^i(z)A^i(z)\big)_{\mu^2}\Big]\Big\{s_A[y,\bar{y}] + (1-\xi)^2 s_A\big[(1-\xi)y,(1-\xi)\bar{y}\big]\Big\}\\
&\quad - \frac{N_c^3}{N_c^2-1}A^i(y-z)A^i(\bar{y}-z)\Big\{s[y,z]\,s[(1-\xi)\bar{y}+\xi z,y]\,s[z,(1-\xi)\bar{y}+\xi z]\\
&\quad + s[z,y]\,s[y,(1-\xi)\bar{y}+\xi z]\,s[(1-\xi)\bar{y}+\xi z,z] + s[\bar{y},(1-\xi)y+\xi z]\,s[(1-\xi)y+\xi z,z]\,s[z,\bar{y}]\\
&\quad + s[(1-\xi)y+\xi z,\bar{y}]\,s[z,(1-\xi)y+\xi z]\,s[\bar{y},z] - \frac{1}{N_c^2}\overline{X}_{y\bar{y}z}\Big\}\Big)\\
&\quad + \frac{1+\xi^2}{1-\xi}\Big(C_F\Big[A^i(y-z)A^i(\bar{y}-z) - \big(A^i(z)A^i(z)\big)_{\mu^2}\Big]\Big\{s_A[y,\bar{y}] + (1-\xi)^2 s\big[(1-\xi)y,(1-\xi)\bar{y}\big]\Big\}\\
&\quad - \frac{N_c}{2}A^i(y-z)A^i(\bar{y}-z)\Big\{s[y,(1-\xi)\bar{y}+\xi z]\,s[z,y] + s[\bar{y},z]\,s[(1-\xi)y+\xi z,\bar{y}]
\end{aligned}
$$

$$
\begin{aligned}
&\quad - \frac{1}{N_c^2}\Big(s[z,(1-\xi)\bar y+\xi z]+s[(1-\xi)y+\xi z,z]\Big)\Big\}\Big)\Big] \\
&- x_p f^g_{\mu^2}(x_p)\int_0^1 d\xi\Big(\frac{1-\xi}{[\xi]_+}+\frac{\xi}{[1-\xi]_+}+\xi(1-\xi)\Big) \\
&\times\Big[C_A\Big[A^i(y-z)A^i(y-z)+A^i(\bar y-z)A^i(\bar y-z)-2\big(A^i(z)A^i(z)\big)_{\mu^2}\Big]s_A[y,\bar y] \\
&\quad+\frac{N_c^3}{2(N_c^2-1)}A^i(y-z)A^i(y-z)\Big\{\frac{1}{N_c^2}\overline{X}'_{y\bar y z} \\
&\qquad-\Big(s[y+\xi(y-z),z+\xi(y-z)]\,s[z+\xi(y-z),\bar y]\,s[\bar y,y+\xi(y-z)] \\
&\qquad+s[z+\xi(y-z),y+\xi(y-z)]\,s[\bar y,z+\xi(y-z)]\,s[y+\xi(y-z),\bar y]\Big)\Big\}+\big(y\leftrightarrow\bar y,\ \overline{X}'\to\overline{X}''\big)\Big]\Big\} \\
&+\frac{g^2}{(2\pi)^3}x_p f^g_{\mu^2}(x_p)\int_0^1\frac{d\xi}{\xi}\int_{y\bar y z}e^{ip_\perp(y-\bar y)}\big[A^i_\xi(y-z)-A^i_\xi(\bar y-z)\big]^2 \\
&\times\Big[\frac{N_c^3}{N_c^2-1}\Big\{s[y,z]\,s[\bar y,y]\,s[z,\bar y]+s[z,y]\,s[y,\bar y]\,s[\bar y,z]-\frac{1}{N_c^2}\overline{X}^A_{y\bar y z}\Big\}-2C_A\,s_A[y,\bar y]\Big]. \qquad \text{(AABKL 4.22)}
\end{aligned}
$$

Their notation:

- $s[x,y]=\frac{1}{N_c}\mathrm{tr}[S^F(x)S^{F\dagger}(y)]$ and $s_A[x,y]=\frac{1}{N_c^2-1}\mathrm{tr}[S^A(x)S^{A\dagger}(y)]$, evolved (BK) to Y_T.
- $X[x,y,z,\bar x,\bar y,\bar z]=\frac{1}{N_c}\mathrm{tr}[S^F(x)S^{F\dagger}(y)S^F(z)S^{F\dagger}(\bar x)S^F(\bar y)S^{F\dagger}(\bar z)]$, with the combinations $\overline{X}_{y\bar y z}$, $\overline{X}'$, $\overline{X}''$, $\overline{X}^A$ of their Eqs. (4.13)–(4.15), (4.18).
- $A^i(y-z)=-\frac{1}{2\pi}\frac{(y-z)^i}{(y-z)^2}$ and $(A^i(z)A^i(z))_{\mu^2}=A^i(z)A^i(z)\theta(z^2\mu^2-1)$ (the collinear subtraction).
- A^i_ξ is A^i times $[1-J_0(|y-z|\sqrt{2\xi x_p P^+/\tau})]$ (Ioffe time τ).
- ξ is the momentum fraction of the *unobserved* gluon. PDFs and FFs are in their scheme, $f_{\mu^2}=f^{\overline{MS}}_{1.26\mu^2}$.
- The $\frac{1+\xi^2}{1-\xi}$ part carries C_F and fundamental dipoles, i.e. it is the quark-initiated channel $q\to g$ (as printed it multiplies f^g; the colour structure requires f^q). Eq. (4.21) has the same layout for quark production.

13 Your result next to AABKL: what matches and what does not

Dictionary.

- $\xi_{\text{AABKL}}=1-\xi_{\text{yours}}$ (unobserved vs. observed fraction).
- $x_p\leftrightarrow\frac{k^+}{\vee}$ (momentum of the produced gluon over the largest projectile momentum); $x_F\leftrightarrow\frac{k^+}{\vee}$ in the hadron convolution.
- $A^i(y-z)\leftrightarrow-\frac{1}{2\pi}\mathcal{K}(y-z)$.
- $s_A[y,\bar y]\leftrightarrow\mathcal{O}$ with the emission kernels of LO.
- $D^g_{H,\mu^2}\leftrightarrow D_{h/g}(\mu_F)$.
- Ioffe time τ, $Y_T\leftrightarrow\Lambda$, $\vee$ cutoffs on p^+.

AABKL (4.20), (4.22)	yours, Eq. (11.1)		comment
hadron convolution $\int_{x_F}^1\frac{d\zeta}{\zeta^2}D(\zeta)\frac{d\bar o}{d^2p}(\frac{p}{\zeta},\frac{x_F}{\zeta})$	$\int_{k^+/\vee}^1\frac{d\zeta}{\zeta^3}D_{h/g}(\zeta)\frac{d^3N_g}{d^3k}(\frac{k^+}{\zeta},\frac{\mathbf{k}}{\zeta})$	same	lower limit from the largest projectile momentum in both
LO: $\frac{1}{(2\pi)^2}xf^g(x)\int e^{ip(y-\bar y)}s_A[y,\bar y]$, order α_s^0	LO: WW gluon of ρ, $\frac{g^2}{2\pi^2k^+}\int X\cdot X'\,\mathcal{O}$, order α_s	differs	collinear gluon (PDF) vs. emission by colour charges ($k_\perp$-dependent)
$\frac{1-\xi}{[\xi]_+}+\frac{\xi}{1-\xi}+\xi(1-\xi)$	$\frac{1}{2N_c}P^{(r)}_{gg,+}(\xi)$	same	with $\xi\to1-\xi$, plus prescription at the soft end in both

AABKL (4.20), (4.22)	yours, Eq. (11.1)		comment
real part $\int_0^{1-x_p} d\xi$	$\int_{k^+/\vee}^1 d\xi$	same	same kinematic limit; both give $\log(1-x_p)$ / $\log(1-\frac{k^+}{\vee})$ through the plus distribution
after the target: $2C_A[A\cdot A-(AA)_{\mu^2}](1-\xi)^2 s_A[(1-\xi)y,(1-\xi)\bar{y}]$	G2 Row IV $\to$ Eq. (7.5), $-\frac{\alpha_s}{2\pi}P^{(r)}_{gg,+}\otimes \mathrm{LO}_\xi[\ell_\Delta]$	same	$\int_z[A\cdot A-(AA)_{\mu^2}]=-\frac{1}{4\pi}\log(r^2\mu^2)$ vs. $-\ell_\Delta=-\log\frac{r^2\mu^2}{c_0^2}$: the scheme shift $c_0^2=4e^{-2\gamma_E}=1.26$ is their $f_{\mu^2}=f^{\overline{MS}}_{1.26\mu^2}$; the rescaled dipole is your $\frac{1}{\xi}$ in the phase
before the target: $2C_A[A\cdot A-(AA)_{\mu^2}]s_A[y,\bar{y}]$	G2 Row I part 2, colour term $ffU(w')U(w)$	kernel differs	same topology; theirs is collinear divergent (PDF subtraction), yours finite: the parent is emitted at $\lvert x-w\rvert$, energy denominators p^+A+k^+B
interference: $\frac{N_c^3}{N_c^2-1}A\cdot A\{sss+\cdots-\frac{1}{N_c^2}\overline{X}\}$	G3 Row III part 2, Eq. (8.8); mixed colour terms of G2 Row I part 2	kernel differs	same topology (parent at $(1-\xi)y+\xi z$ vs. your $y'=\frac{k^+w'+p^+z}{P}$)
virtual: $-x_p f\int_0^1 d\xi(\dots)\{C_A[AA+AA-2(AA)_{\mu^2}]s_A+\frac{N_c^3}{2(N_c^2-1)}AA[\frac{\overline{X}'}{N_c^2}-sss]\}$	G1 Row V 2a, Eq. (8.5): $-ffU(x)U(z)+N_cU(x)$; G1 Rows II, V δ-terms	colour same, kinematics differ	theirs $\int_0^1$ with no x_p limit; your Row V has $\int_0^{k^+}$ (no $\vee$), but Row II has $\xi\geq\frac{k^+}{\vee}\Rightarrow$ the remainder (7.3)
last term: $\int\frac{d\xi}{\xi}[A_\xi-A_\xi]^2\{\frac{N_c^3}{N_c^2-1}(sss+sss-\frac{\overline{X}^A}{N_c^2})-2C_A s_A\}$	$\mathcal{N}\log\frac{\vee}{\Lambda}\mathcal{J}$	colour same, regularization differs	both are the real−virtual soft-gluon (BK/JIMWLK) structure; theirs finite with the Ioffe time ("not extra evolution"), yours a fixed interval $\log\frac{\vee}{\Lambda}$
$\frac{1+\xi^2}{1-\xi}C_F$ term; Eq. (4.21)	—	absent	quark channels; $n_f=0$ and no quarks in your projectile
—	running coupling, Eq. (6.3)	absent	AABKL: LO is α_s^0, nothing to renormalize
—	Eq. (7.4)	absent	needs an emission vertex in LO
—	linear polarization, Eq. (7.6)	absent	their incoming gluon is unpolarized; yours is polarized along the WW field
—	parent emitted after the target by the rotated charge (G2 Row I part 2; G3 Rows V, VI)	absent	their parent always exists before the target
—	three-ρ, four-ρ terms	absent	single parton in AABKL
—	remainder, Eq. (7.3)	absent	all collinear poles are absorbed in AABKL

In words.

- **What matches** is everything that comes from the splitting of a gluon that can be treated as on shell:
  - the splitting function, with the same plus prescription and the same exact kinematic limit ($1-x_p\leftrightarrow 1-\frac{k^+}{\vee}$, including the $\log(1-\cdot)$ it produces);
  - the after-target term, which in both is the fragmentation-function term, with the same scheme conversion ($1.26=c_0^2$);
  - the hadron convolution;
  - the colour structures of the before-target, interference and virtual terms;
  - the soft real−virtual structure of their last term, which is your evolution term.
- **What does not match** is everything that comes from your projectile being colour charges:
  - LO is $O(\alpha_s)$ with a $k_\perp$-dependent emission;

- the before-target and interference kernels are finite and contain the emission distance $|x-w|$, so there is no PDF subtraction;
- the parent gluon is linearly polarized;
- the parent can be emitted after the target;
- the coupling of LO must be renormalized (β_0 cancels, the constant $\frac{67}{9}-\frac{\pi^2}{3}$ remains);
- there are three- and four-ρ terms.

- **Different regularization, same physics:** Ioffe time vs. p^+ cutoffs. Their last term is finite because of A_ξ; your $\log\frac{\vee}{\Lambda}\mathcal{J}$ is a fixed interval.
- **One mismatch that is a problem on your side:** their virtual term carries no projectile-momentum limit and all poles go into PDFs and FFs. Your Group I Row II carries $\xi\geq\frac{k^+}{\vee}$, and with $\vee$ exact this leaves the pole of Eq. (7.3).

14 Questions

Q1 The pole of Eq. (7.3).

- With $\vee$ exact, $\frac{\alpha_s}{2\pi}\frac{2}{\epsilon}N_c\log(1-\frac{k^+}{\vee})$LO is left over. It comes from $\log(\frac{\vee}{k^+}-1)$ in Group I Row II part 1, i.e. the limit $\xi\geq\frac{k^+}{\vee}$ (parent momentum $\leq\vee$) of $|\Psi^B_{g\rho}\rangle$.
- Is that limit intended for the virtual one-ρ correction? In AABKL the corresponding virtual term is $\int_0^1 d\xi$.
- Or is a real term with the same limit missing?
- Until this is settled, the result is finite only up to $O(\frac{k^+}{\vee})$.

Q2 Four ρ's. In Groups II and III (and Group I Row IV) the four-ρ terms run to $\vee-k^+$, in Group I Rows I, III to $\vee$. Exactly, this leaves Eq. (9.1). With your kinematics ($p^+<\vee$ for the independently emitted gluon) all ranges would be $(\Lambda,\vee)$ and Eq. (9.1) would vanish. Which is intended?

Q3 Real rows. The file uses $p^+<\vee-k^+$, i.e. parent momentum $P^+=p^++k^+<\vee$. Your statement "$\vee>p^+,k^+>\Lambda$" alone allows p^+ up to $\vee$. If $P^+<\vee$ is not intended, every $\vee-k^+$ and $\log(1-\frac{k^+}{\vee})$ of the real rows changes.

Q4 Keep the symmetrized three-ρ finite terms (first saturation correction), or drop them (Gaussian projectile)?

Q5 Do you want the commutator two-ρ terms of the three-ρ rows written out row by row?

Q6 Group I Row I′ (the window $|p^+-k^+|<\Lambda$): is its finite part included anywhere?

Q7 Scheme: $d_\perp=2-\epsilon$ (your W) or 2? It decides ℓ_Δ vs. $\ell_\Delta+1$ in Eq. (7.5).

Q8 Observable: gluons or hadrons? Eq. (11.1) is for hadrons; the gluon spectrum alone is not collinear safe.

Q9 Coupling in LO: $\alpha_s(\mu_F)$, or KLSZ's $g(c_0^2/X^2)g(c_0^2/X'^2)$?

Q10 Confirm the measure of Group III Row III part 2 (F4).

Q11 Should AABKL's Ioffe-time restriction be imposed instead of the p^+ cutoffs? It would change the finite terms at $O(1)$.

15 Checks

Scripts in `nlo_gluon_paper/finite/`:

- `ft.py`: your file, transcribed;
- `verify.py`: Table 1;
- `simp_exact.py`: Sec. 8, → `simp_exact.json`;
- `checks_exact.py`: tallies and four ρ, → `checks_exact.json`;
- `checks.py`: three-ρ relations, → `checks.json`.

check	result
file vs. exact p^+ integrals, closed-form rows	spread $\leq 7 \times 10^{-13}$
file vs. exact, plus-prescription rows (lower limit 0)	spread $\leq 1 \times 10^{-4} = O(\Lambda/k^+)$
Sec. 8 closed-form rows ($\Lambda = 0$) vs. file at $\Lambda = 10^{-10}$, $\vee/k^+ = 1.3, 2.5, 10, 300$	$\leq 1 \times 10^{-9} = O(\Lambda/k^+)$
Sec. 8 E_1 forms (G1 V(1), G3 III(1), III(2), V, VI) vs. quadrature of the file	$\leq 1 \times 10^{-14}$
G1 V part 2, Eq. (8.4), vs. quadrature of the file	$\leq 3 \times 10^{-9}$
master integrals $\mathcal{M}_{0,1,2}$ vs. quadrature	3×10^{-11}
Eqs. (8.7), (8.8) as written vs. code	6×10^{-16}
pole tally, $\delta(1-\xi)$ coefficient, exact in a (`sympy`)	$11/6 - \log(1-a)$
β_0 in Rows II + V; finite $\hat{E}$ terms vs. Eqs. (6.3), (7.4), (7.3)	difference 0; difference 0
four ρ, symmetrized: all lines / lines with range $\vee - k^+$	max. $\lvert\Sigma\rvert$ 3×10^{-14} (terms up to 221) / max. $\lvert\Sigma\rvert = 12.3$–$24.3$ (terms up to 163)
G1 Row IV + G3 Row I (integrand); G2 Row I(1)A + G3 Row II	6×10^{-15}; 4×10^{-16}
$U(z)$ network, G1 Row III vs. Eq. (10.1) (does not cancel)	0.16–8.6